## 1 Propose candidates

SAM 3, frozen



image

"Locate the handle of the mug."

phrases $\phi(x)$

the handle of the mug

the mug

**SAM 3 · point grid**
32×32, query-free, ≤ 256

**SAM 3 · text**
top 3 per phrase

### Candidate pool

N = 260




grid masks  SAM 3 text instances (#rank)

### Containment tree



$$\text{parent}(v) = \arg\min_u |m_u| \;\text{ s.t. }\; |m_u \cap m_v| \geq 0.8\,|m_v|$$

e.g. mug ⊃ handle; used between mask tokens

## 2 Mask tokenization

candidate $m_v$



**Appearance** $a_v = \sum_j \tilde{c}_{v,j}\, e_j$
what it shows: the MLLM's image tokens under the mask

**Geometry** $g_v \in \mathbb{R}^9$
where it is: box, area, depth in the tree

**Source** $q_v \in \mathbb{R}^5$
where it came from: grid or SAM 3 text (rank, score)

concatenate

**MLP + RMS norm**

$$\hat{x}_v = \text{MLP}\big([\,a_v;\; g_v;\; q_v\,]\big)$$

position (mRoPE): mask centroid



like a word token: content + position + type

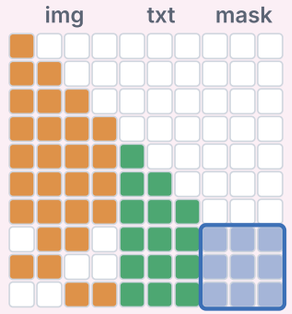
## 3 The MLLM selects

Qwen3-VL-4B + LoRA




**Large language model**
Qwen3-VL-4B + LoRA, 64.2M trainable parameters

### Region-restricted attention (every layer)




- a mask token sees only its own image region, plus the query
- image and query tokens never see the mask tokens
- between mask tokens: a learned bias $b_{\ell,\,\text{rel}(u,v)}$ by tree relation

**Selection head**

$$s_v = \lambda\,\cos\big(\psi_q(h_{\langle\text{SEL}\rangle}),\, \psi_k(z_v)\big) + \beta \qquad v^\star = \arg\max_v\, s_v$$

scores over the pool




answer $m_{v^\star}$

